Roll No.	10013	Dept.	EE	Section	A7	

Instructions:

1. This question paper contains a total of **1** page (**1** side of paper).
2. Write your name, roll number, department, and section on every side of every sheet of this booklet
3. Write final answers neatly with a blue/black pen in the given boxes.
4. **Answers written outside the box will NOT be graded.**

Total **10 Marks**

Q. 1: Write the output of the following program in the appropriate box and answer the question. (2+2+2 = 6 Marks)

```
1   #include<stdio.h>
2   void func1(int *arr){
3     for(int i=1; i<=5;i++)
4       (i-1)[arr] = (i%5);
5   }
6   void func2 (int a[1] , int arr){
7     a[0] = arr + 3;
8   }
9  int main(){
10   int arr[6] = {2,2,2,2,2};
11   func1(arr+1);
12   for(int i=0; i<6; i++)
13           printf("%d ", i[arr] );
14   printf("\n");
15   func2(arr, *arr);
16   for(int i=0; i<6; i++)
17           printf("%d ", i[arr] );
18   return 0;
19 }
```

Output
2 1 2 3 4 0
5 1 2 3 4 0

Will the answer change if we write func2 as:

```
void func2 (int a[1] , int arr){
  int temp = arr + 4;
  a = &temp;
}
```

Explain your answer

yes,
By func2,
'a' will now contain memory location of 'temp'.
The actual array wasn't updated at all.

Q. 2: Write the output of the following sequence of instructions **long arr2[90];** **int arr1[80], a1=(&arr2[60]), a2=(&arr2[10]), b1=(&arr1[60]), b2=(&arr1[10]);** **printf("%d %d", a1-a2 , b1-b2);** Explain your answer. (2+2 Marks)

output : 400 200
as size of (long int) = 8
size of (int) = 4

→ let &arr2[0] = x
then &arr2[10] = $x + 8*10 = a_2$
&arr2[60] = $x + 8*60 = a_1$
$a_1 - a_2 = 8*50 = 400$, similarly $b_1 - b_2 = 200$